Faisal Ahmed

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Google Scholar | *GitHub*

Research Interest

- IoT systems, edge AI, and IoT security, tested on real hardware.

Education

Bachelor of Science in Computer Science and Engineering **May/2026**
East West University, Dhaka 1212

- Result CGPA 3.73/4.00
- Capstone research: PAUMIoT, supervised by Hasan Mahmood Aminul Islam; published at DCOSS-IoT 2026.

Higher Secondary School Certificate (HSC) **January/2021**
Dhaka College, New Market, Dhaka 1205

- GPA 5.00/5.00 in all subjects. Average : 96%

Research Experience

Capestone Research, IoT Systems **2025 – Present**
East West University, Dhaka. Supervised by Hasan Mahmood Aminul Islam

- Built the ESP32 edge AI client for PAUMIoT that switches between MQTT, CoAP and HTTP at runtime
- Wrote a safety shield that blocks unsafe protocol choices on low battery or weak signal
- Set up the ESP32 and sensor testbed and ran all experiments; P99 latency 185 ms vs. 509 ms for Mosquitto
- Built the communication layer and core algorithms of SEFIoT; tested it with 20,000 emulated sensors
- Extended The ONE simulator with a CCN layer and a publication register to measure stale content

Independent Research, IoT Attack Measurement **Apr 2026 – Present**
Honeypot study; journal paper in preparation

- Deployed Cowrie, Glutton, Dionaea and OpenCanary honeypots in Frankfurt
- Collected 3.9 million attack events from 28,963 IPs over 112 days
- Gathered Shodan and Censys data on 51,738 IoT devices and malware data from four threat feeds
- Clustered attackers into campaigns

Publications

Conference papers

1. H. M. A. Islam, M. M. R. Maharaz, **Faisal Ahmed**, M. F. H. Dipto, M. M. H. Shafin, and M. Georgiades, "Towards a protocol-agnostic universal middleware for IoT with edge intelligence," in *Proc. 22nd Int. Conf. Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, 2026, doi: *10.1109/DCOSS-IoT69657.2026.00046*.
2. M. M. Rahman, **Faisal Ahmed**, M. F. H. Dipto, M. M. H. Shafin, S. M. N. Shahriar, P. Akibuzzaman, and H. M. A. Islam, "Emulating sensor profile towards IoT simulator development," in *Proc. IEEE 2nd Int. Conf. Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, doi: *10.1109/QPAIN69676.2026.11546398*.
3. F. R. Sams, S. K. Supti, M. J. Ali, **Faisal Ahmed**, M. F. H. Dipto, R. Gani, M. B. N. Shuci, M. R. A. Rashid, and S. H. Ripon, "Interpretable machine learning for early diabetes prediction with statistical validation and explainable AI," in *Proc. IEEE Int. Conf. Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2025, doi: *10.1109/BECITHCON69222.2025.11504208*.

Submitted:

1. M. M. Rahman, **Faisal Ahmed**, M. F. H. Dipto, and M. M. H. Shafin, "Freshness before the cache: Measuring content staleness at ingress in CCN over delay-tolerant networks," submitted to *13th Int. Conf. Next Generation Computing, Communication, Systems and Security (NSysS)*, 2026.

Journal papers

In Preparation:

1. **Faisal Ahmed**, M. M. R. Maharaz, M. F. H. Dipto, M. M. H. Shafin, and M. Georgiades, "PAUMIoT: Safety-shielded, federated, cost-aware transport adaptation for heterogeneous IoT."
2. **Faisal Ahmed**, N. Yeasmin, R. Hossen, and M. Georgiades, "Passive attribution of role-separated IoT attack ecosystems through orthogonal behavioral fingerprints."

Skills

- **Hardware and embedded:** ESP32 (FreeRTOS), sensors, hardware-in-the-loop testing
- **Security and measurement:** Cowrie, Dionaea, Glutton, OpenCanary, Shodan, Censys, threat-intelligence feeds
- **Languages:** C, Go, Python, C#
- **Systems and data:** Linux servers, PostgreSQL, The ONE simulator

Teaching Experience

Instructor **01/10/23 - 01/03/24**
Step to Backend, Online sessions

- Taught the fundamentals of backend systems. Instructed on database implementation and deployment.

Honors and Awards

- Medha Lalon(Dean's) Fund Scholarship
- Talent pool Scholarship in PEC and JSC
- Dhaka Reporters Unity Result Excellency Award given by former **Education Minister Nurul Islam Nahid**

Experience

Co-Founder & Software Engineer **25/07/2025 - Present**
Trelet, Dhaka, Bangladesh

- Built a product variant engine (Go, PostgreSQL, Next.js) with N-dimensional attributes, tiered pricing and stock-aware filtering; fixed a Cartesian-product query bug.
- Implemented end-to-end Stripe payments with signed webhook verification and a 25+ case adversarial test suite covering webhook tampering and payment-intent attacks (100% pass).
- Built a three-tier RBAC system with 23 security tests; found and removed a critical static-secret authentication bypass during a security audit.

Full-stack Developer (Remote) **01/10/2025 - 31/12/2025**
RN Jobforce, Shropshire, England

- Built Django backend services, including WebSocket live chat with sub-100ms latency and Celery-based notifications.
- Developed an OpenAI-powered interview module with dynamic question generation and real-time candidate feedback.

Software Engineer **01/04/2025 - 30/09/2025**
Progress IT Solution, Dhaka, Bangladesh

- Built a C# video processor with FFmpeg, 15 custom shaders, and OpenCV frame interpolation.
- Developed a WPF/WinForms app automating domain registration and GPO management for a cybersecurity client.